5: CONTINUOUS RANDOM VARIABLES

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Litterature:

H. H. Hamedani, *Probability and Statistics: The Science of Uncertainty*, 3rd ed., https://www.probabilitycourse.com

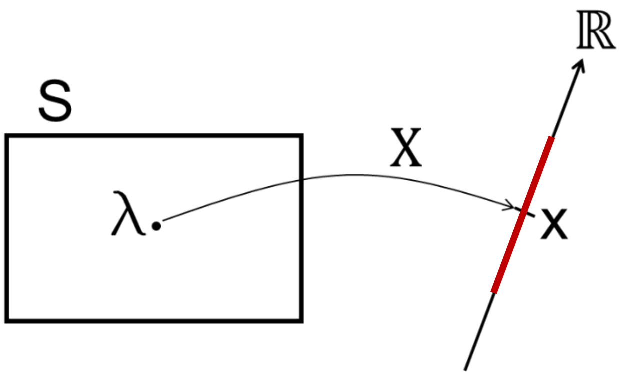
Course: Stokastisk modellering og behandling (SWSTS-01)

Continuous Random Variable

A **continuous random variable** is a real-valued function

$$X : S \to \mathbb{R}$$

that assigns a **real number** to each outcome $\lambda \in S$ of a random experiment, such that the set of possible values $X(\lambda)$ lies in a **continuous interval** of real numbers.



Example: Examine a randomly selected patient

S={all patients present in the room}.

Each outcome $\lambda_i \in S$ is a patient selected.

Define X by $X(\lambda_i) = blood\ pressure\ of\ patient\ i$

Range

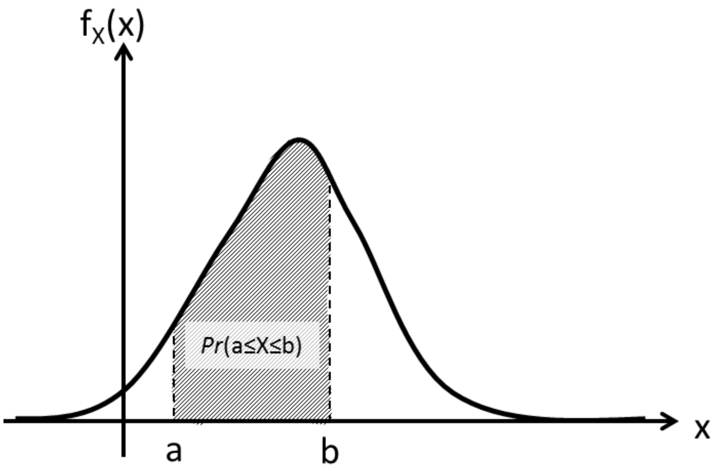
Blood pressure can take any real value in a physiological interval, e.g. $X \in [0, \infty)$

Probability Density Function (PDF)

Let X be a continuous random variable. Then the **probability density function** (pdf) $f_X(x)$ is defined such that:

$$\Pr(a \leq X \leq b) = \int_a^b f_X(x)\,dx$$

This gives the probability that X lies within the interval $[a, b]$.



Key Properties of the PDF:

- **Non-negativity**:

$$f_X(x) \geq 0 \quad \text{for all } x$$

- **Normalization**:

$$\int_{-\infty}^{\infty} f_X(x)\,dx = 1$$

(The total area under the curve must be 1.)

Probability Density Function (PDF)

Important Consequences:

- The probability of any exact value is **zero**:

$$\Pr(X = x) = 0$$

- Intervals of any kind give the same probability:

$$\Pr(a < X < b) = \Pr(a \le X \le b) = \Pr(a \le X < b)$$

- It is possible for the **value** of $f_X(x)$ to be **greater than 1**. This is okay as long as the **area** under the curve is 1.

Mass = Probability and Density = Probability pr unit lenght

In physics, suppose mass is distributed along a thin rod.

We define **mass density** at position x as

$$\rho(x) = \frac{\text{mass in small interval}}{\text{length of that interval}}.$$

More precisely,

$$\text{mass in } [x, x + \Delta x] \approx \rho(x)\,\Delta x$$

So:

- $\rho(x)$ is **not** mass
- $\rho(x)$ is **mass per unit length**

To get actual mass, you integrate:

$$\text{mass on } [a, b] = \int_a^b \rho(x)\,dx.$$

Probability Density Function (PDF)

Example: Exponential PDF

Let X be a continuous random variable with probability density function (PDF):

$$f_X(x) = \begin{cases} ce^{-x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

where c is a positive constant.

What c-value makes this $f_X(x)$ a valid PDF?

Use normalization:

$$\int_{-\infty}^{\infty} f_X(x)\, dx = 1$$

Since $f_X(x) = 0$ for $x < 0$, we compute:

$$\int_0^{\infty} ce^{-x}\, dx = c\int_0^{\infty} e^{-x}\, dx = c\left[-e^{-x}\right]_0^{\infty} = c(0 - (-1)) = c$$

Set this equal to 1:

$$c = 1$$

Quiz

A continuous random variable X has the following probability density function (pdf):

$$f_X(x) = \begin{cases} \frac{1}{2}e^{-\frac{1}{2}x}, & \text{for } x \geq 0 \\ 0, & \text{for } x < 0 \end{cases}$$

What is the probability that $X = 2$?

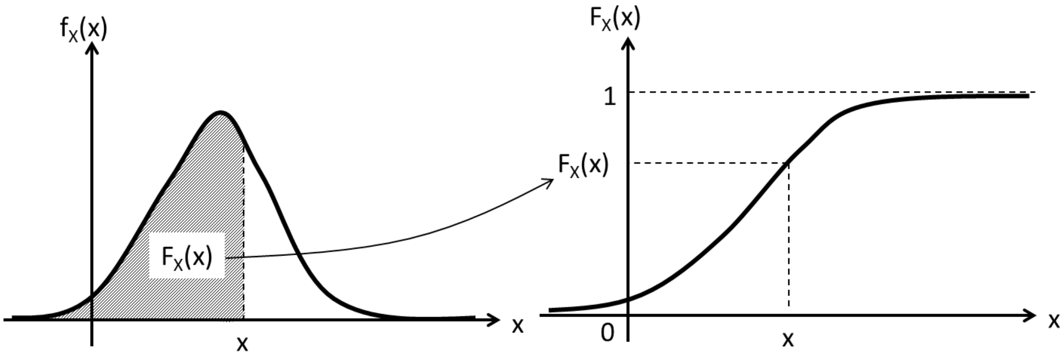
A: 0 **B:** $\frac{1}{2}e^{-1}$ **C:** $\frac{1}{2}$ **D:** 2

Cumulative Distribution Function (CDF)

The **cumulative distribution function** of a continuous random variable X is defined as:

$$F_X(x) = \Pr(X \leq x) = \int_{-\infty}^{x} f_X(u)\,du$$

- This represents the total probability accumulated from $-\infty$ up to x.



Key Properties of CDFs:

- $0 \leq F_X(x) \leq 1$ (because probabilities range from 0 to 1)
- $F_X(x)$ is **non-decreasing** and **continuous**
- For any interval $[a, b]$:

$$\Pr(a \leq X \leq b) = \int_a^b f_X(x)\,dx = F_X(b) - F_X(a)$$

- For tail probability:

$$\Pr(X > x) = 1 - \Pr(X \leq x) = 1 - F_X(x)$$

Cumulative Distribution Function (CDF)

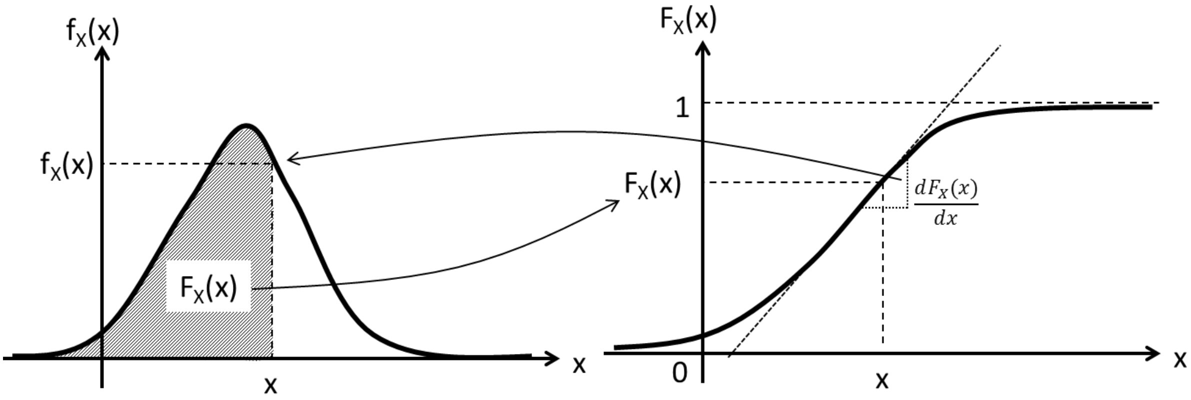
Relationship Between PDF and CDF

- **From PDF to CDF:**

$$F_X(x) = \int_{-\infty}^{x} f_X(u)\,du$$

- **From CDF to PDF** (if $F_X(x)$ is differentiable):

$$f_X(x) = \frac{d}{dx} F_X(x)$$



Cumulative Distribution Function (CDF)

Example: Find CDF from PDF

$$f_X(x) = \begin{cases} e^{-x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

We compute it piecewise:

- If $x < 0$, then $f_X(u) = 0$ for all $u < x$, so:

$$F_X(x) = 0$$

- If $x \geq 0$, then:

$$F_X(x) = \int_0^x e^{-u}\, du = \left[-e^{-u}\right]_0^x = 1 - e^{-x}$$

Thus, the CDF is:

$$F_X(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-x}, & x \geq 0 \end{cases}$$

Quiz

A continuous random variable X has the following cumulative distribution function (CDF):

$$F_X(x) = \begin{cases} 1 - e^{-\frac{1}{2}x}, & \text{for } x \geq 0 \\ 0, & \text{for } x < 0 \end{cases}$$

What is the **probability density function** $f_X(x)$ for the random variable when $x \geq 0$?

A: $-e^{-\frac{1}{2}x}$

B: $\frac{1}{2}e^{-\frac{1}{2}x}$

C: $\frac{1}{2}$

Expectation

For **continuous random variables**, sums become integrals and PMFs become PDFs:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x\, f_X(x)\, dx$$

Example: Expected Value of a Continuous Random Variable

Let X be a continuous random variable with probability density function (PDF):

$$f_X(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

We want to compute the expected value $\mathbb{E}[X]$.

Solution:

Use the formula for the expected value of a continuous random variable:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx$$

Since $f_X(x) = 0$ outside the interval $[0, 1]$, we simplify:

$$\mathbb{E}[X] = \int_0^1 x \cdot 2x\, dx = \int_0^1 2x^2\, dx$$

Now compute the integral:

$$\int_0^1 2x^2\, dx = 2 \cdot \frac{x^3}{3}\Big|_0^1 = \frac{2}{3}$$

Expectation

LOTUS (expectation of a function of a random variable):

Continuous Case (replace the sum and PMF with integral and PDF):

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx$$

Example

Let X be a continuous random variable with probability density function (PDF):

$$f_X(x) = \begin{cases} x + \frac{1}{2}, & \text{for } 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

We are asked to find the expected value of X^n, that is, $\mathbb{E}[X^n]$, where $n \in \mathbb{N}$.

Solution:

$$\mathbb{E}[X^n] = \int_{-\infty}^{\infty} x^n f_X(x)\, dx$$

$$= \int_0^1 x^n \left(x + \frac{1}{2}\right) dx$$

$$= \left[\frac{1}{n+2} x^{n+2} + \frac{1}{2(n+1)} x^{n+1}\right]_0^1$$

$$= \frac{3n+4}{2(n+1)(n+2)}.$$

Expectation

Linearity of Expectation:

(Valid for both discrete and continuous random variables)

- **For any constants** $a, b \in \mathbb{R}$ and any random variable X:

$$\mathbb{E}[aX + b] = a \cdot \mathbb{E}[X] + b$$

- **For any finite collection of random variables** $X_1, X_2, \ldots, X_n$:

$$\mathbb{E}[X_1 + X_2 + \cdots + X_n] = \mathbb{E}[X_1] + \mathbb{E}[X_2] + \cdots + \mathbb{E}[X_n]$$

Proof of first part: (use LOTUS)

$$\mathbb{E}[aX + b] = \int_{-\infty}^{\infty} (ax + b) f_X(x)\, dx.$$

Split the integral:

$$\mathbb{E}[aX + b] = a \int_{-\infty}^{\infty} x f_X(x)\, dx + b \int_{-\infty}^{\infty} f_X(x)\, dx.$$

But

$$\int_{-\infty}^{\infty} f_X(x)\, dx = 1.$$

Therefore

$$\mathbb{E}[aX + b] = a \int_{-\infty}^{\infty} x f_X(x)\, dx + b.$$

Variance

The **variance** of a random variable X is defined as the expected squared deviation from its mean:

$$\operatorname{Var}(X) = \mathbb{E}[(X - \mu_X)^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Equivalent formula by use of LOTUS:

$$\operatorname{Var}(X) = \int_{-\infty}^{\infty} (x - \mu_X)^2 f_X(x)\, dx$$

- Here, $\mu_X = \mathbb{E}[X]$

Example: Find the mean and variance of a continuous random variable

Let X be a continuous random variable with PDF:

$$f_X(x) = \begin{cases} \frac{3}{x^4}, & x \geq 1 \\ 0, & \text{otherwise} \end{cases}$$

Step 1: Compute the mean $\mathbb{E}[X]$

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx = \int_{1}^{\infty} x \cdot \frac{3}{x^4} dx = \int_{1}^{\infty} \frac{3}{x^3} dx = \left[-\frac{3}{2x^2} \right]_1^{\infty} = \frac{3}{2}$$

Step 2: Compute $\mathbb{E}[X^2]$

$$\mathbb{E}[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x)\, dx = \int_{1}^{\infty} x^2 \cdot \frac{3}{x^4} dx = \int_{1}^{\infty} \frac{3}{x^2} dx = \left[-\frac{3}{x} \right]_1^{\infty} = 3$$

Step 3: Compute the variance

$$\operatorname{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 3 - \left(\frac{3}{2}\right)^2 = 3 - \frac{9}{4} = \frac{3}{4}$$

Variance

Properties of Variance:

(Also valid for both discrete and continuous random variables)

1. **Not a linear operator**

$$\mathrm{Var}(aX + b) = a^2 \cdot \mathrm{Var}(X)$$

- Scaling by a changes the variance.
- Shifting by b has no effect.

2. **Additivity over independent random variables**

   If $X = X_1 + X_2 + \cdots + X_n$ and the X_i are independent:

$$\mathrm{Var}(X) = \sum_{i=1}^{n} \mathrm{Var}(X_i)$$

Discrete vs. Continuous Random Variables

Discrete (Σ)	Continuous ($\int$)
CDF: $F_X(x) = \Pr(X \leq x) = \sum_{x_i \leq x} f_X(x_i)$ non-decreasing step-function	CDF: $F_X(x) = \Pr(X \leq x) = \int_{-\infty}^{x} f_X(x)dx$ non-decreasing continuous function
PMF: $f_X(x) = F_X(x) - F_X(x-1)$	PDF: $f_X(x) = \frac{dF_X(x)}{dx}$
$E[X] = EX = \mu_X = \bar{X} = \sum_{x_k \in R_X} x_k \cdot f_X(x_k)$	$E[X] = EX = \mu_X = \bar{X} = \int_{-\infty}^{\infty} x \cdot f_X(x)dx$
$E[g(X)] = \sum_{x_k \in R_X} g(x_k) \cdot f_X(x_k)$	$E[g(X)] = \int_{-\infty}^{\infty} g(x) \cdot f_X(x)dx$
$Var(X) = \sigma_X^2 = \sum_{i=1}^{n} (x_i - \bar{x})^2 f_X(x_i)$ $= E[X^2] - E[X]^2$	$Var(\mathrm{X}) = \sigma_X^2 = \int_{-\infty}^{\infty} (x - \bar{x})^2 \cdot f_X(x)dx$ $= E[X^2] - E[X]^2$

Special Distributions

Normal (Gaussian) Distribution

VERY important!!

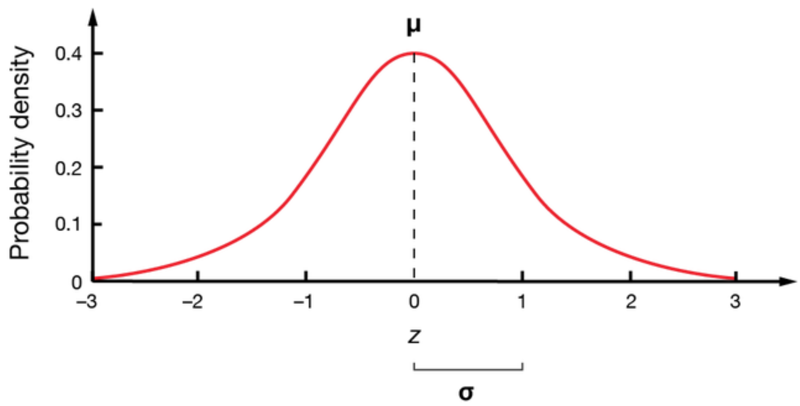
Standard Normal Distribution**:**

- Mean: $\mu = 0$
- Variance: $\sigma^2 = 1$

A random variable $Z \sim \mathcal{N}(0,1)$ has the following:

- **PDF** (probability density function):

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right)$$



- **CDF** (cumulative distribution function):

$$F_Z(z) = \Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right) du$$

NOTE: No closed expression for $\Phi(z)$. tabulated in **tables** and available in software like **Python** and **MATLAB**.

Special Distributions

To show that the **standard normal distribution** $Z \sim \mathcal{N}(0,1)$ has

$$\mathbb{E}[Z] = 0, \qquad \mathrm{Var}(Z) = 1,$$

we use only the **definitions**

$$\mathbb{E}[Z] = \int_{-\infty}^{\infty} z f_Z(z)\, dz, \qquad \mathrm{Var}(Z) = \mathbb{E}[Z^2] - \big(\mathbb{E}[Z]\big)^2$$

Insert the PDF:

$$\mathbb{E}[Z] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z e^{-z^2/2}\, dz$$

- $e^{-z^2/2}$ is **even**:

$$e^{-(-z)^2/2} = e^{-z^2/2}.$$

- z is **odd**:

$$(-z) = -z.$$

So the integrand

$$ze^{-z^2/2}$$

is **odd** × **even** = **odd**.

Hence

$$\int_{-\infty}^{\infty} (\text{odd function})\, dz = 0.$$

Now one can show that $Var(x) = E[Z^2] - 0^2 = E[Z^2] = 1$

Transformations of random variables

The Method of Transformations

If X is a random variable and you define a new one by

$$Y = g(X),$$

then Y is random too. The goal is to find the **distribution** of Y:

Transform the CDF of X:

$$F_Y(y) = \Pr(Y \leq y) = \Pr(g(X) \leq y) = \Pr\big(X \leq g^{-1}(y)\big) = F_X\big(g^{-1}(y)\big)$$

Nb. In order for g to have **inverse**, it needs to be **injective** - fx. strictly increasing or decreasing (but the inequality flips if g is strictly decreasing.)

Find PDF of Y:

$$f_Y(y) = \frac{d}{dy}F_Y(y) = \frac{d}{dy}F_X(g^{-1}(y)) = F_X'\big(g^{-1}(y)\big) \cdot \frac{d}{dy}g^{-1}(y)$$

Nb. g needs to be **differentiable**

Transformations of random variables

Linear transformations

Let

$$Y = aX + b, \qquad a \neq 0.$$

Case 1: $a > 0$

$$F_Y(y) = \Pr(aX + b \leq y) = \Pr\left(X \leq \frac{y-b}{a}\right) = F_X\left(\frac{y-b}{a}\right).$$

We now differentiate:

$$F_Y(y) = F_X\left(\frac{y-b}{a}\right)$$

apply the **chain rule**:

$$\frac{d}{dy} F_X(u(y)) = F_X'(u(y)) \cdot u'(y)$$

Where $u(y) = \frac{y-b}{a}$

- $F_X'(x) = f_X(x)$
- $u'(y) = \frac{1}{a}$

So

$$\boxed{f_Y(y) = \frac{1}{a} f_X\left(\frac{y-b}{a}\right), \qquad a > 0}$$

Case 2: $a < 0$

The inequality flips: $F_Y(y) = 1 - F_X\left(\frac{y-b}{a}\right) \Rightarrow f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$

Transformations of random variables

Mean/variance of linear transformation

$$\mathbb{E}[Y] = a\,\mathbb{E}[X] + b, \qquad \mathrm{Var}(Y) = a^2\,\mathrm{Var}(X).$$

Linear transformation on Standard Normal distributed random variable:

Start with

$$Z \sim \mathcal{N}(0,1).$$

Where

$$\mathbb{E}[Z] = 0, \qquad \mathrm{Var}(Z) = 1.$$

Now consider a linear transformation

$$X = aZ + b.$$

$b = \mu$ is the new mean value

$$\mathbb{E}[X] = \mathbb{E}[aZ + b] = a\,\mathbb{E}[Z] + b = a \cdot 0 + b = b.$$

$$\mathrm{Var}(X) = \mathrm{Var}(aZ + b) = a^2\,\mathrm{Var}(Z) = a^2 \cdot 1 = a^2.$$

$a^2 = \sigma^2$ is the new Variance value

We can obtain a **general normal distribution** $X \sim \mathcal{N}(\mu, \sigma^2)$ by applying a **linear transformation** to $Z \sim \mathcal{N}(0,1)$:

$$X = \sigma Z + \mu$$

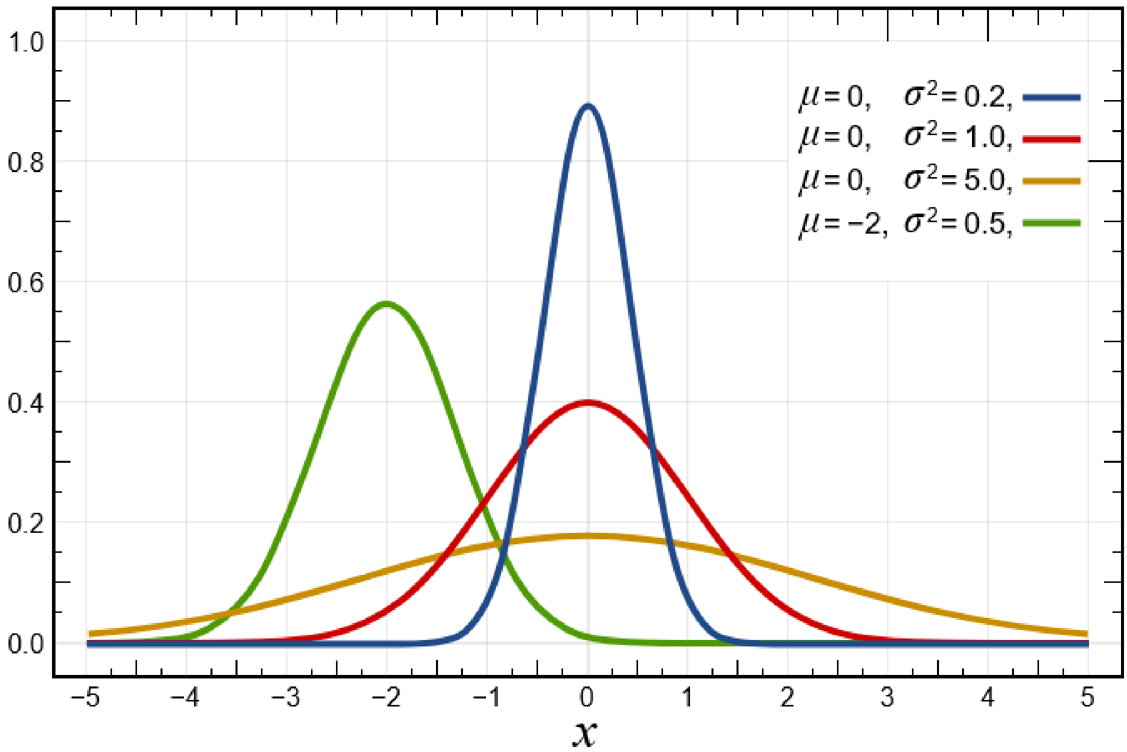
Transformations of random variables

Linear transformation of standard normal PDF:

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2},$$

Apply the linear transformation

$$f_X(x) = \frac{1}{\sigma} \cdot f_Z\left(\frac{x-\mu}{\sigma}\right)^2 = \frac{1}{\sigma} \cdot \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$$



Transformations of random variables

Linear transformation of standard normal CDF:

Starting from

$$X = \sigma Z + \mu,$$

solve for Z:

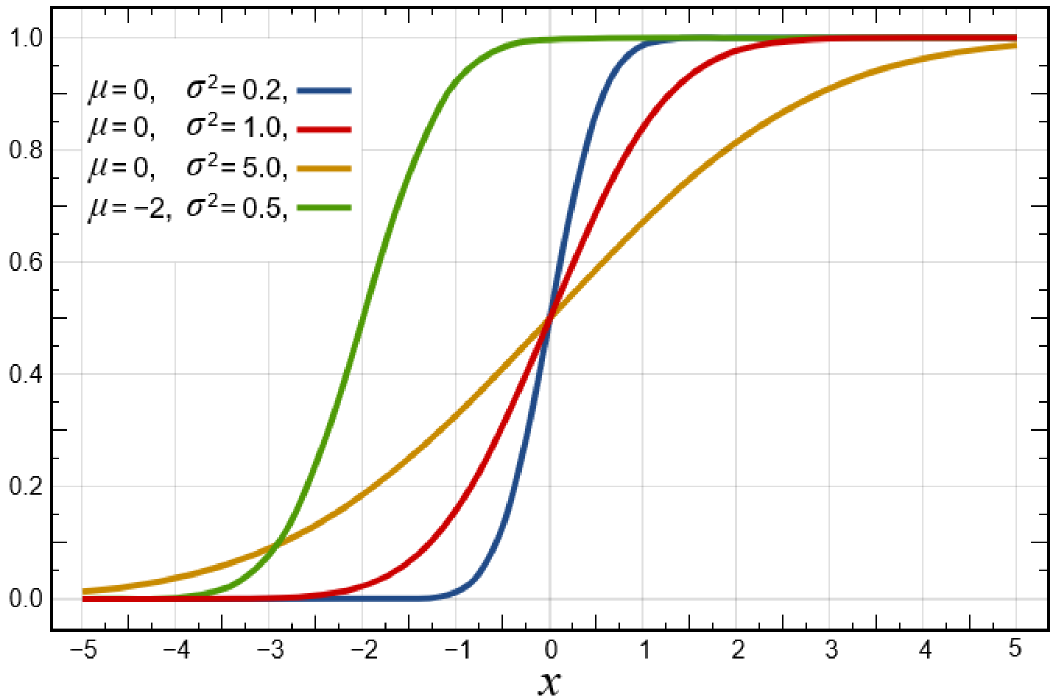
$$Z = \frac{X - \mu}{\sigma}.$$

Then

$$\Pr(X \leq x) = \Pr(\sigma Z + \mu \leq x) = \Pr\left(Z \leq \frac{x - \mu}{\sigma}\right) = \Phi\left(\frac{x - \mu}{\sigma}\right).$$

Standard normal CDF given by:

$$\Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right) du$$



Transformations of random variables

Example: Using **Matlab** and **Python** for CDF calculations

Let $X \sim \mathcal{N}(10, 2^2)$.

Find $P(X \leq 9)$ using both the general CDF and a z-transformation.

```python
from scipy.stats import norm

# Directly with general parameters
prob1 = norm.cdf(x=9, loc=10, scale=2)      # ≈ 0.31

# Using standard normal
z = (9 - 10) / 2
prob2 = norm.cdf(z)                         # ≈ 0.31
```

Result:

$$P(X \leq 9) = \Phi(-0.5) \approx 0.31$$

Transformations of random variables

Example: Compute PDF Value

Let $X \sim \mathcal{N}(10, 2^2)$. Find the value of the **PDF** at $x = 9$.

Manual Calculation

$$f_X(9) = \frac{1}{2\sqrt{2\pi}} \exp\left(-\frac{(9-10)^2}{2\cdot 2^2}\right) = \frac{1}{2\sqrt{2\pi}} \exp\left(-\frac{1}{8}\right) \approx 0.176$$

python

```python
from scipy.stats import norm

# PDF using general normal parameters
prob = norm.pdf(x=9, loc=10, scale=2)
print(prob)    # ≈ 0.176
```

The PDF value at $x = 9$ tells us the height of the bell curve at that point – **NOT** the probability!

Transformations of random variables

Example: Compute probability by use of CDF

Let $X \sim \mathcal{N}(10, 2^2)$.

Find the probability that X lies between 8 and 12:

Method 1: Using the CDF

The cumulative distribution function gives:

$$P(8 < X < 12) = F_X(12) - F_X(8) = \Phi\left(\frac{12-10}{2}\right) - \Phi\left(\frac{8-10}{2}\right)$$

$$= \Phi(1) - \Phi(-1) \approx 0.8413 - 0.1587 = 0.6826$$

```python
from scipy.stats import norm

# Parameters
mu = 10
sigma = 2

# CDF values at bounds
p1 = norm.cdf(12, loc=mu, scale=sigma)
p2 = norm.cdf(8, loc=mu, scale=sigma)

# Probability between 8 and 12
prob = p1 - p2
print(prob)  # ≈ 0.6826
```

Transformations of random variables

Example: nonlinear transformation

Given

$$F_X(x) = \begin{cases} 0, & x \le 0, \\ x^4, & 0 < x \le 1, \\ 1, & x \ge 1. \end{cases}$$

And let $Y = \frac{1}{X}$

Step 1: Range of Y:

as $X \to 0^+$, $Y \to \infty$. Also $X = 1 \Rightarrow Y = 1$.

$Y = \frac{1}{X} \in [1, \infty)$

Step 2: Transform the CDF

$$F_Y(y) = \Pr(Y \le y) = \Pr\left(\frac{1}{X} \le y\right) = \Pr\left(X \ge \frac{1}{y}\right) = 1 - F_X\left(\frac{1}{y}\right) = 1 - \frac{1}{y^4}$$

Step 3: Differentiate to get the PDF

$$f_Y(y) = \frac{d}{dy}\left(1 - y^{-4}\right) = 4y^{-5} = \frac{4}{y^5}.$$

Quick check

$$\int_1^\infty \frac{4}{y^5}\,dy = 4\left[\frac{y^{-4}}{-4}\right]_1^\infty = \left[-y^{-4}\right]_1^\infty = 0 - (-1) = 1.$$

Special Distributions

Uniform Distribution

Let $X \sim \text{Uniform}(a, b)$, meaning that X is a continuous random variable uniformly distributed over the interval $[a, b]$. Its **probability density function (PDF)** is:

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & \text{for } a < x < b \\ 0, & \text{otherwise} \end{cases}$$

Expectation value:

$$\mathbb{E}[X] = \int_a^b x \cdot \frac{1}{b-a}\, dx$$

Now compute:

$$\mathbb{E}[X] = \frac{1}{b-a} \int_a^b x\, dx = \frac{1}{b-a} \left[\frac{1}{2}x^2\right]_a^b = \frac{1}{b-a} \cdot \left(\frac{1}{2}b^2 - \frac{1}{2}a^2\right)$$

$$= \frac{1}{2(b-a)}(b^2 - a^2) = \frac{1}{2(b-a)}(b-a)(b+a) = \frac{a+b}{2}$$

To find the variance, we can find EX^2 using LOTUS:

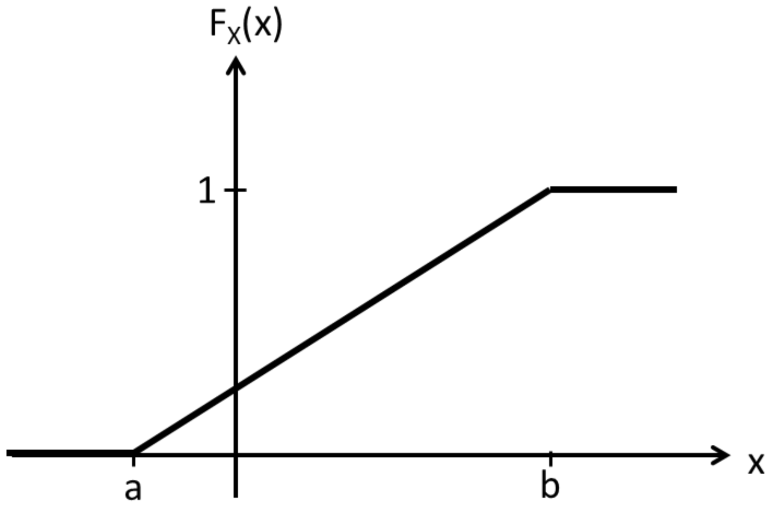
$$EX^2 = \int_{-\infty}^{\infty} x^2 f_X(x)\, dx = \int_a^b x^2 \left(\frac{1}{b-a}\right) dx = \frac{a^2 + ab + b^2}{3}$$

Therefore,

$$\text{Var}(X) = EX^2 - (EX)^2 = \frac{(b-a)^2}{12}$$

Quiz

A continuous random variable X has the following CDF



How is X distributed?

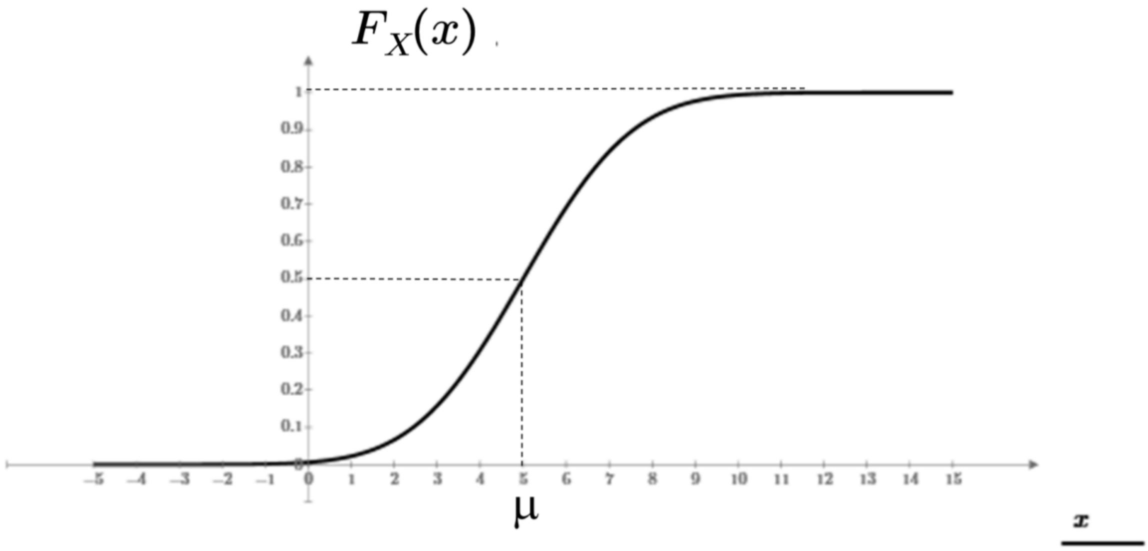
A: $\mathcal{B}inomial(8, -2)$

B: $\mathcal{U}(-2, 8)$

C: $\mathcal{N}(3, 5)$

Quiz

A continuous random variable X has the following CDF



How is X distributed?

A: $\mathcal{B}inomial(8, -2)$

B: $\mathcal{U}(-2, 8)$

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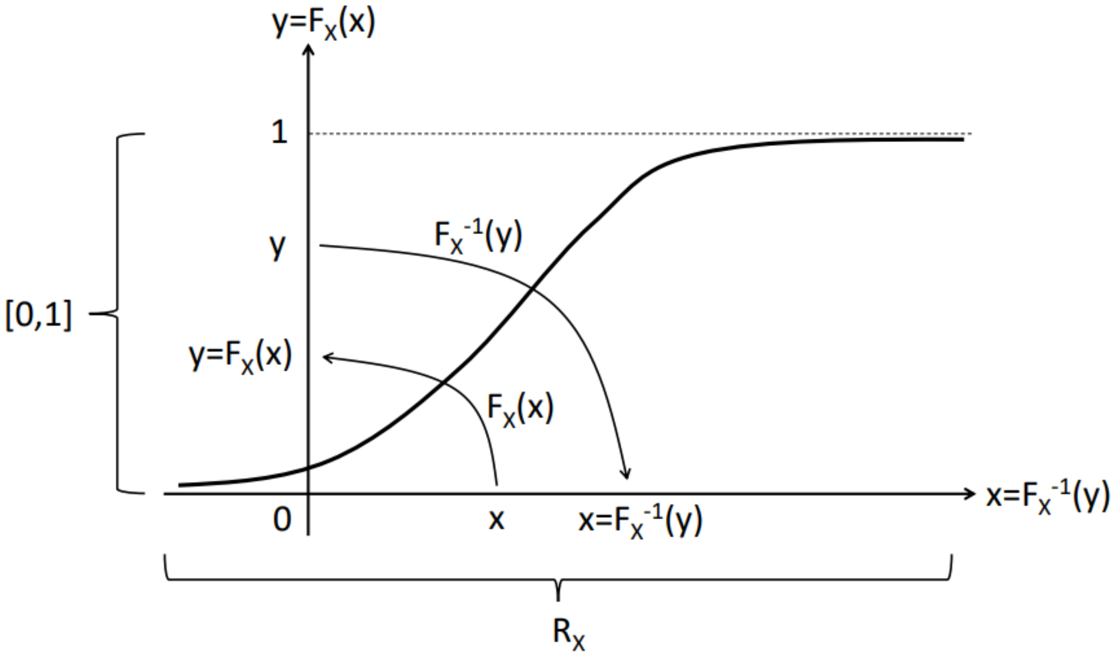
Sampling From Any Distribution

For test or simulation you need testdata ("measurements") randomly sampled from a given distribution:



For test or simulation you need testdata ("measurements") randomly sampled from a given distribution:

- Standard distributions: $X \sim \mathcal{N}(\mu, \sigma^2) \rightarrow randn()$
  $X \sim \mathcal{U}(a, b) \rightarrow rand()$
- Other distributions:
  - Find the cdf of the distribution: $F_X(x)$
  - Find the inverse of the cdf: $y = F_X(x) \;\Rightarrow\; x = F_X^{-1}(y)$
  - Draw a random sample: $y \sim \mathcal{U}[0;1]$
  - Insert into the inverse cdf: $x = F_X^{-1}(y)$
  - The samples $X = x$ is distributed according to: $F_X(x)$

Sampling From Any Distribution

Example

For exponential distribution with rate λ, the **CDF** is:

$$F_X(x) = 1 - e^{-\lambda x}, \quad x \geq 0$$

To sample x, invert the CDF:

$$y = 1 - e^{-\lambda x} \quad \Rightarrow \quad x = -\frac{1}{\lambda}\ln(1-y), \quad y \sim U[0,1]$$

python

```python
import numpy as np
import matplotlib.pyplot as plt

# Parameters
lambda_ = 1
N = 10000  # number of samples

# Step 1: Generate uniform samples
y = np.random.uniform(0, 1, N)

# Step 2: Inverse transform sampling
x = -np.log(1 - y) / lambda_

# Plot histogram
plt.hist(x, bins=50, density=True, edgecolor='black')
plt.title('Exponential Sampling using Inverse Transform')
plt.xlabel('x')
plt.ylabel('Density')
plt.grid(True)
plt.show()
```